

STATEMENT OF REQUIREMENT

FOR

3D Printed Group 1 UAS Kits

1. General Information

New requirement(s) to purchase 3D printed Group 1 (20lbs or less) small Unmanned Aerial System (sUAS) kits and ground control stations associated support equipment as defined in the requirements below. The requirement includes the digital files to print our own Group 1 UAS, build the UAS with the parts and components kits associated with the UAS model purchased. The requirement will be for the initial purchase of a complete sUAS build kit which gives the ORNG the opportunity to 3D print the airframe, build out the printed airframe with the vendor provided components, and repair Group 1 UAS in austere and tactical environments without heavily relying on logistical or supply chains, i.e., FEDEX or Amazon Prime. The Contractor will furnish all National Defense Authorization Act (NDAA) required critical components, other UAS components, and 3D print files for fabricating the UAS airframe necessary to provide the identified products in Paragraph 2 to the 249th Regional Training Institute, Hermiston, Oregon.

2. Baseline Requirements Standards (24 Kits – Airframe kits & Ground Control Stations)

2.1. CLIN0001: Group 1 UAS Airframe Kit and 3D Print File for Airframe:

2.1.1. Completed airframe with all flight components and battery weight:1-20lbs.

2.1.2. Airframe design must be modular in nature.

2.1.2.1. All flight hardware components essential to the control and operation of the airframe, as an assembly, must be capable of tool-less replacement or reconfiguration in 1 minute or less in a field environment, exclusive of the ground control station.

2.1.2.2. All propulsion hardware essential for lift and mobility of the airframe, as an assembly, must be capable of tool-less replacement or reconfiguration in 1 minute or less in a field environment, exclusive of the ground control station.

2.1.2.3. All firmware parameters essential to the intended nominal operation of the airframe must be capable of reconfiguration in 1 minute or less in a field environment.
Reference 2.2.11

2.1.2.4. Airframe fuselage/chassis must be capable of being additively manufactured from Composite ABS Glass Fiber-reinforced filament or greater for the purpose of end-user build, rapid repair, and customization of the airframe as required by mission parameters.

2.1.2.5. Airframe motor arms must be capable of folding for packing the UAS by Soldier in backpack.

2.1.2.6. Airframe motor arms must feature positive retention mechanisms to securely lock the motor arms in their deployed, flight ready condition.

2.1.2.7. Airframe must be able to have MIL-STD-1913 picatinny rail attached, minimum of 2" in length but no longer than 4" to readily accept external end-user payloads and/or attachments.

- 2.1.2.8. Airframe must provide a means of controlling external end-user payload functions and attachments via the ground control station (GCS), with an interface similar to but not limited to Pulse-Width Modulation (PWM).
- 2.1.3. Airframe is capable of mission profile Intelligence, Surveillance and Reconnaissance (ISR): Airframe must be capable of maneuvers at a speed of 45 Knots with a 1kg payload with an endurance equal to or greater than 30 minutes of flight and possess the ability to upgrade/reconfigure in three minutes or less in a field environment for the following additional flight profiles:
 - 2.1.3.1. High Speed Attack: Airframe must be capable of maneuvers at speeds of 70 Knots with a 1kg payload with endurance equal to or greater than 15 minutes of flight.
 - 2.1.3.2. Heavy Lift: Airframe must be capable of maneuvers at a speed of 35 Knots with a 3kg payload with an endurance equal to or greater than 20 minutes of flight.
- 2.1.4. Airframe operates with electric propulsion only.
- 2.1.5. Flight times at cruise speed in ISR configuration equal to or greater than 30 minutes.
 - 2.1.5.1. Nominal cruise speed in ISR configuration equal to or greater than 30 kts.
 - 2.1.5.2. Maximum cruise speed in ISR configuration equal to or greater than 44 kts.
 - 2.1.5.3. Minimum cruise speed in ISR configuration equal to or greater than 21 kts.
- 2.1.6. ISR configuration service ceiling (above sea level) up to 12,000 ft.
- 2.1.7. Red, Green, Blue (RGB) Camera Sensor equal to daylight camera with available room to add/integrate thermal and/or infrared camera(s) on a separate purchase.
- 2.1.8. RGB Camera specifications: Equal or greater than 11.9MP @ 24fps (4608 x 2592 pixels), global or rolling shutter, 66 degrees horizontal field of view (HFOV), 10.0x digital zoom.
- 2.1.9. IR Camera specifications: Resolution equal or greater than 24fps, 35 degree horizontal field of view (HFOV), shutterless, 2.0x digital zoom.
- 2.1.10. Live Streaming; Equal or greater than 720p @ 24fps (RGB).
- 2.1.11. Video data captures equal to or greater than recorded image/video format of H265 MP4 video file.
- 2.1.12. Static frame capture equal to or greater than still images of 11.9MP (4K Resolution).
- 2.1.13. Camera gimbal mechanism capable of rotation on the pitch axis (Up/Down).
- 2.1.14. Camera sensor Gimbal rotation speed adjustable up to 360 degrees per second.
- 2.1.15. Support ISM band frequencies equal to 900mHz, 1W max and 2.4GHz, video, 1W max, for radio communications.

- 2.1.16. Command/control range equal to or greater than 10km or 6.2 miles (line of sight).
- 2.1.17. Digital Video range equal to or greater than 5km using directional antenna and 2km using omnidirectional antenna.
- 2.1.18. Critical components are NDAA compliant. Critical components, as defined by Headquarters, Department of the Army (HQDA) EXORD 272-25, consist of flight controllers, radios, data transmission devices, gimbals, ground control systems, and operating software.
- 2.1.19. Additional Components:
 - 2.1.19.1. Quantity: 1 set of replacement propellers.
 - 2.1.19.2. Quantity: 2 Lithium-Ion batteries.
 - 2.1.19.3. Quantity: 1 Lithium-Ion battery charger.
- 2.1.20. Complete 3D print files for airframe to the end-user for printing, building, repairing, and customization (Airframes to be printed by 249 RTI).
- 2.1.21. Water resistant equal to or greater than IP54.

2.2. CLIN0002: sUAS Ground Control Station (GCS) for Airframe:

- 2.2.1. Weight equal to or less than 1 kg.
- 2.2.2. Touch screen equal to or greater than 8 inches.
- 2.2.3. Battery life equal to or greater than 2 hours with internal battery.
- 2.2.4. Operating system compatible with Android 11.
- 2.2.5. Software updates equal to or greater than Over the Air (OTA) capabilities.
- 2.2.6. Support ISM band frequencies equal to 900mHz, 1W max and 2.4GHz, video, 1W max, for radio communications.
- 2.2.7. Critical components NDAA Compliant.
- 2.2.8. Water resistant equal to or greater than IP54.
- 2.2.9. GCS must provide a means of controlling external end-user payloads and attachments and their corresponding functions (e.g., drop mechanisms), equal to or greater than PWM.
- 2.2.10. GCS is configured to transmit and receive a telemetry and data link capable of the intended and nominal control of the UAS, offline mission planning, payload control, and video streaming.

2.2.11. GCS is capable of receiving, reviewing, and transmitting flight controller parameters in order to reconfigure the UAS to meet various mission profiles.

2.2.11.1. High Speed Attack mission profile.

2.2.11.2. Intelligence, Surveillance and Reconnaissance (ISR) mission profile.

2.2.11.3. Heavy Lift mission profile.

3. Delivery, Inspection and Acceptance

3.1. Place of Delivery. 249 RTI, 78080 Ordnance Road, Bldg. 5004, Hermiston, OR 97838

3.2. Inspection and acceptance will occur at time of delivery by a government representative. A comprehensive bill of lading or delivery receipt shall accompany deliverables that details, at minimum, all serial numbers associated with the Ground Control Station (GCS).

4. Government Furnished Equipment (GFE) or Property (GFP)

5. Documentation

5.1. The Contractor shall provide all owner's manuals and specification sheets and all parts.

5.2. The Contractor shall provide a serial number for each assembly build kit.

5.3. The Contractor shall provide a serial number each ground control station.

5.4. The Contractor shall provide the necessary files for 3D printing of the airframes.

5.5. The Contractor shall provide the necessary assembly and operation instructions for the airframes and synchronization instructions for use with the ground control stations.

6. Glossary

6.1. Field Environment - an operational or training setting located outside of permanent military installations (garrisons) where service members conduct missions or exercises under conditions that simulate or constitute actual combat.